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Calculations of magnetic hyperfine structure constants for the low-lying rovibrational levels of LiH, HF, CH^+, and BH

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Abstract

Spin–rotation constants for the six lowest vibrational states and the first excited end-over-end rotational state were calculated for several isotopomers of the $X\,^1\Sigma^+$ diatomic molecules: LiH, HF, CH^+ and BH. For the ground vibrational state, the rotational dependence of the spin–rotation constant was also investigated and was found to be very weak. The electronic calculations were done at various levels of the polarization propagator approximation using atomic basis sets adapted for calculations of magnetic properties. The calculated spin–rotation constants for the isotopomers of LiH and HF, namely their rovibrational dependence, are in good agreement with the available experimental data. The extremely large values of the spin–rotation constants of C and B in CH^+ and BH previously predicted by coupled perturbed Hartree–Fock calculations at the equilibrium internuclear distances were confirm by the present correlated and rovibrational averaged calculations. For the ground vibrational and first rotational state the spin–rotation constant of B in ^{11}BH in $^{11}CH^+$ are predicted to be -459.70 and -1186.15 kHz, respectively. The results are discussed in the light of recent calculations of the nuclear quadrupole coupling constants, since the relative strengths of both the magnetic and electric parameters determined the resulting pattern of the hyperfine structure of molecular rovibrational spectra. Improved experimental values for the nuclear magnetic shielding constants of HF ($\sigma_F = 418.0 \pm 2.0$, $\sigma_H = 28.6 \pm 2.0$) at the equilibrium geometry were derived from recent measurements and calculated diamagnetic contributions.

1. Introduction

Interactions between nuclear electric quadrupole or magnetic dipole moments and the electric or magnetic fields generated by the electrons in a rotating molecule give rise to the hyperfine structure of the molecular spectra. Their measurement and the analysis provide valuable information about molecular properties and molecular structure (see e.g. Refs. [1–4]). This role is even more emphasized by recent advances in laser induced fluorescence and Doppler-free saturated absorption spectroscopies which provide a wealth of hyperfine data for various molecular systems and cover a still wider field of rovibrational quantum numbers (e.g. Refs. [5–8]).

The variations of the hyperfine structure pattern with the rotational and vibrational quantum numbers, observed in molecular spectra, reflect the dependence of the relevant molecular properties on the nuclear geometry. Their study gives therefore a

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deeper insight into the character of the molecular bond and the nature of the perturbations arising from the coupling of the ground electronic wit excited electronic states [4]. Reflecting the growing importance of theoretical investigations of hyperfine structure parameters in excited molecular rovibrational states, Vojtík et al. published recently a series of calculations of the rovibrational dependence of the nuclear quadrupole coupling constant of diatomic molecules: H_2, LiH, LiH^+, BH, CH^+ [9–13]. Although the nuclear quadrupole couplings give rise to dominating and most frequently observed hyperfine interaction, the magnetic dipole terms in the hyperfine structure Hamiltonian [14], i.e. the spin–rotation coupling, and the direct and indirect nuclear spin–spin interactions have also to be considered in any analysis of the hyperfine splitting. The relative size of all hyperfine coupling constants determines the coupling scheme in each rovibrational level [5,15]. From the three magnetic parameters, the direct spin–spin coupling constants can be calculated directly from the molecular geometry. On the other hand, the spin–rotation and direct spin–spin coupling constants contain electronic contributions which can be expressed as second-order sum-over-states properties and require ab initio calculations [16,17]. Since for $^1\Sigma$ states the latter is orders of magnitude smaller and its effect on the hyperfine structure, if any, is only secondary, we concentrate here on the theoretical investigation of the rovibrational dependence of the spin–rotation constants and direct nuclear spin–spin coupling constants.

Our approach is based on the Born–Oppenheimer separation of the electronic and nuclear motion and consists of ab initio electronic calculations of the electronic part of the spin–rotation constant as a function of the internuclear distance, and a rovibrational averaging of the property curve obtained as a sum of the electronic and corresponding nuclear terms. The electronic part of the spin–rotation constant, a second-order sum-over-states property [18], can be calculated by polarization propagator methods [16,19]. These methods have been applied to the calculation of electric [20] and magnetic [21,22] second-order properties for the equilibrium geometry. Gaussian basis sets for first row atoms, which give molecular magnetic properties converged at correlated levels of the polarization propagator approximation (PPA), have been developed and can be employed in this work. In our recent studies of the magnetic properties, including the spin–rotation constants, of NH_3 [23] and H_3O^+ [24] effective magnetic constants were evaluated within the framework of the non-rigid inverter Hamiltonian from property surfaces obtained by polarization propagator calculations.

We present calculations of the spin–rotation constants and direct nuclear spin–spin coupling constants and their rovibrational dependence for some isotopomers of the diatomic $^1\Sigma$ molecules: LiH, HF, CH^+ and BH. They are calculated for the six lowest vibrational states, with $J = 1$, for several rotational levels of the vibrational ground state. These magnetic hyperfine structure data complement the results of the recent calculations of nuclear quadrupole coupling constants [10,11,13].

Our results for LiH can be compared with experimental values for several rovibrational states obtained from molecular beam electric resonance spectra [25–28]. Also the molecular beam electric resonance spectra of $H^{19}F$ and $D^{19}F$ have been measured several times and from their analysis the hyperfine structure parameters for several rovibrational states have been determined with high accuracy [3,29,30]. These experimental data have made the HF molecule a test case for the development of theoretical approaches. The spin–rotation constant of HF and its rovibrational dependence is also important since the experimental absolute scale of the nuclear magnetic shielding constants σ of F is based on it. The equilibrium values for the nuclear magnetic shielding constants are reproduced by state-of-the-art theoretical methods [31,32]. Their rovibrational dependence and the rovibrational dependence of the corresponding spin–rotation constants, however, had found much less attention.

There are no experimental data available for any isotopomer of BH or CH^+. However, Fowler and Steiner [33] recently obtained extremely large values for the spin–rotation constants of B and C in a coupled perturbed Hartree–Fock calculation (CHF). This implies that the hyperfine structure pattern of the rotational spectra of these molecules should be of interest for experimentalists. The large values of the boron quadrupole coupling constants [11] ensure a dominant role for the quadrupole coupling in the

hyperfine structure of all isotopic species of BH and the magnetic interaction could be observed as additional splittings of the lines. The hyperfine structure of the $^{13}CH^+$ rovibrational spectra, on the other hand, arises entirely from magnetic interaction, and could be observed because of the large value of the spin–rotation constant of C. The study of Fišer and Vojtík [13] drew our attention to the radioactive isotopomer $^{11}CH^+$ and $^{11}CD^+$ for which the hyperfine splitting originates from a coupling of the electric and magnetic interactions of comparable strengths. The variations of the coupling scheme [15] with the isotopic species will be discussed by comparison of the quadrupole and magnetic hyperfine constants.

In the following sections we give details about the calculations and present the spin–rotation and direct spin–spin interaction constants of the isotopomers of LiH, HF, CH^+ and BH.

2. Theory and computational details

The terms corresponding to the spin–rotation, the direct spin–spin and the nuclear quadrupole interactions in the hyperfine structure Hamiltonian are [15,34]

$$H = -h\sum_K I_K \boldsymbol{M}^K J + h\sum_{K\neq L} I_K \boldsymbol{D}^{KL} I_L + h\sum_K \frac{e\boldsymbol{Q}_K \boldsymbol{q}^K}{h}, \quad (1)$$

where I_K and I_L are dimensionless spin operators of the nuclei K and L, J is the dimensionless molecular end-over-end rotation operator, $\boldsymbol{M}^K$ is the spin–rotation coupling tensor of nucleus K, $\boldsymbol{D}^{KL}$ is the direct nuclear spin–spin coupling tensor of nuclei K and L, $\boldsymbol{Q}_K$ is the nuclear electric quadrupole moment tensor of nucleus K and $\boldsymbol{q}^K$ is the electric field gradient at the nucleus K. The summations in Eq. (1) are over all nuclei.

In diatomic molecules, with the nuclei along the z axis, the only non-zero elements of the spin–rotation tensor $\boldsymbol{M}^K$ are its xx and yy components [18], $(\boldsymbol{M}^K)_{xx} = (\boldsymbol{M}^K)_{yy} = M^K$. M^K is then called the spin–rotation constant of nucleus K. It can be written for a closed shell molecule [18] as

$$M^K(R) = \frac{g_K}{2\pi I(R)}\left(\frac{hm_e}{2\pi m_p}\sigma^p_{K,\perp}(R) + \frac{e\mu_0\mu_N}{4\pi}\frac{Z_L}{R}\right), \quad (2)$$

where g_K is the g-factor of nucleus K, $I(R)$ is the moment of inertia of the diatomic molecule corresponding to the internuclear distance R, $\sigma^p_{K,\perp}(R)$ is the perpendicular component of the paramagnetic contribution to the nuclear magnetic shielding tensor of nucleus K calculated with the gauge origin on the nucleus K and Z_L is the charge of the other nucleus. $\sigma^p_K(R)$ in Eq. (2) is the only quantity which cannot be obtained from the molecular geometry alone. As a second-order property it can be expressed as a sum over excited electronic states:

$$\boldsymbol{\sigma}^p_K(R) = \frac{e^2}{2m^2}\frac{\mu_0}{4\pi}\sum_{n\neq 0}\frac{1}{E_0 - E_n} \times\left(\langle 0|\sum_i \frac{\boldsymbol{l}_i(R_K)}{r^3_{iK}}|n\rangle\langle n|\sum_i \boldsymbol{l}_i(R_K)|0\rangle + \langle 0|\sum_i \boldsymbol{l}_i(R_K)|n\rangle\langle n|\sum_i \frac{\boldsymbol{l}_i(R_K)}{r^3_{iK}}|0\rangle\right), \quad (3)$$

where $\boldsymbol{l}_i(R_K)$ is the angular momentum of electron i with respect to the position of nucleus K, r_{iK} is the distance between the nucleus K and the electron i, $\{|n\rangle\}$ is a complete set of excited electronic states with energies $\{E_n\}$, $|0\rangle$ and E_0 correspond to the ground state. Sum-over-states expressions like Eq. (3) can be evaluated as polarization propagators [19]. Four levels of approximation to the polarization propagator are used in this work : the uncorrelated first-order or random-phase approximation (RPA) which is for static properties equivalent to the coupled perturbed Hartree–Fock method (CHF), the second-order polarization propagator approximation (SOPPA) [35], and two coupled cluster approximations based on reference states for either doubles or singles and doubles (CCDPPA and CCSDPPA) [36].

The calculations were performed with the Odense version of the RPAC 9.1 program package [37]. The corresponding $M^K(R)$ values were then calculated from expression (2).

The direct nuclear spin–spin coupling tensor is a symmetric traceless Cartesian tensor [15]

$$D_{ij}^{KL} = \frac{\mu_0}{4\pi} \frac{\mu_N^2 g_K g_L}{h} \frac{r_{KL}^2 \delta_{ij} - 3 r_i r_j}{r_{KL}^5}, \tag{4}$$

which, for a diatomic molecule with internuclear distance R, is fully determined by the constant $D^{KL}(R)$

$$D^{KL}(R) = \frac{\mu_0}{4\pi} \frac{\mu_N^2 g_K g_L}{h R^3}. \tag{5}$$

For the analysis of the hyperfine structure of the rotation spectra of diatomic molecules the spin–rotation, direct nuclear spin–spin coupling and the nuclear quadrupole coupling constants corresponding to the v and J states, $M_{v,J}^K$, $D_{v,J}^{KL}$ and $eQ_K q_{v,J}^K/h$, are necessary. In the framework of the Born–Oppenheimer approximation the spin–rotation constant $M^K(R)$, direct nuclear spin–spin coupling constant $D^{KL}(R)$ and the nuclear quadrupole coupling constant $eQ_K q^K(R)/h$ are first calculated as a function of the internuclear distance. The constants for the vth vibrational and Jth end-over-end rotational state can then be evaluated as a expectation values of $M^K(R)$, $D^{KL}(R)$ and $eQ_K q^K(R)/h$ over the radial part of the rovibrational wavefunction $\chi_{v,J}$

$$\begin{aligned} M_{v,J}^K &= \langle \chi_{v,J} | M^K(R) | \chi_{v,J} \rangle_R, \\ D_{v,J}^{KL} &= \langle \chi_{v,J} | D^{KL}(R) | \chi_{v,J} \rangle_R, \\ q_{v,J}^K &= \langle \chi_{v,J} | q^K(R) | \chi_{v,J} \rangle_R. \end{aligned} \tag{6}$$

The function $\chi_{v,J}$ is the solution of the one-dimensional Schrödinger equation

$$-\frac{h^2}{8\pi^2\mu} \frac{\mathrm{d}^2\chi_{v,J}}{\mathrm{d}R^2} + \left(U(R) + \frac{h^2}{8\pi^2\mu} \frac{J(J+1)}{R^2} - E_{v,J} \right) \chi_{v,J}(R) = 0. \tag{7}$$

Here, μ is the reduced mass of the two nuclei and $U(R)$ the Born–Oppenheimer electronic potential energy. Eq. (7) is solved numerically by the Cooley–Numerov integration technique [38,39].

3. Results and discussion

3.1. LiH

The rotational hyperfine structure of lithium hydride has been very extensively investigated by molecular beam resonance spectroscopy [25–28]. These measurements of the hyperfine structure constants in two vibrational states can be compared with our calculated spin–rotation constants as can their calculated dependence on the vibrational state.

The electronic calculations of σ^p for LiH were done at various levels of the polarization propagator approximation (PPA) with a (13s8p6d) contracted to [8s8p6d] Gaussian basis set for Li and (7s3p) contracted to [5s3p] Gaussian basis set for H. The hydrogen basis, denoted sp, has been used previous in the calculation of magnetic properties [21,22]. The Li basis set is derived from the basis set of Roos and Sadlej [40] by removing the f functions and adding

Table 1
Dependence of σ^p (in ppm) and M (in kHz) of LiH on the level of the polarization propagator approximation for the equilibrium bond length ($R_e = 3.015\, a_0$)

			^{7}LiH		^{6}LiD	
	σ_{Li}^p	σ_H^p	M^{Li}	M^H	M^{Li}	M^D
RPA	−18.23	−12.94	−9.86	9.71	−2.18	0.87
SOPPA	−18.50	−12.94	−10.08	9.71	−2.23	0.87
CCDPPA	−17.96	−12.92	−9.65	9.75	−2.13	0.87
CCSDPPA	−17.54	−12.95	−9.31	9.69	−2.06	0.87

two sets of p functions with the exponents $\xi_p = 33.0$ and $\xi_p = 100.0$. The effect of additional polarization functions was investigated, but it was found that they did not significantly alter the results of the correlated PPA calculations. The paramagnetic contributions to the nuclear magnetic shielding constants $\sigma^p_{Li}(R)$ and $\sigma^p_H(R)$ were calculated for 11 internuclear distances (R) in the range between $1.5\,a_0$ and $4\,a_0$. The equilibrium bond length values of $\sigma^p_{Li}(R_e)$ and of $\sigma^p_H(R_e)$ and of the corresponding spin–rotation constants for ^{7}LiH and ^{6}LiD, $M^{Li}(R_e)$ and $M^{H/D}(R_e)$, obtained at various levels of the PPA are given in Table 1. The paramagnetic contribution to the shielding constant of hydrogen and thus the spin–rotation constants $M^{H/D}$ are almost unaffected by electron correlation over the whole range of geometries, whilst the absolute value of σ^p_{Li} decreases by $\sim 4\%$. Our RPA results are very close to the extended basis set CHF calculation of Keil and Ahlrichs [41]. Fig. 1 shows the dependence of the RPA and CCSDPPA results for σ^p_H and of σ^p_{Li} on the internuclear distance R. As the σ^p_H curves are virtually identical and the $\sigma^p_{Li}(R)$ curves, although different, are parallel within the chosen interval of R, the rovibrational corrections are independent on the level of approximation and we therefore consider only the results of the highest level (CCSDPPA) for the rovibrational averaging.

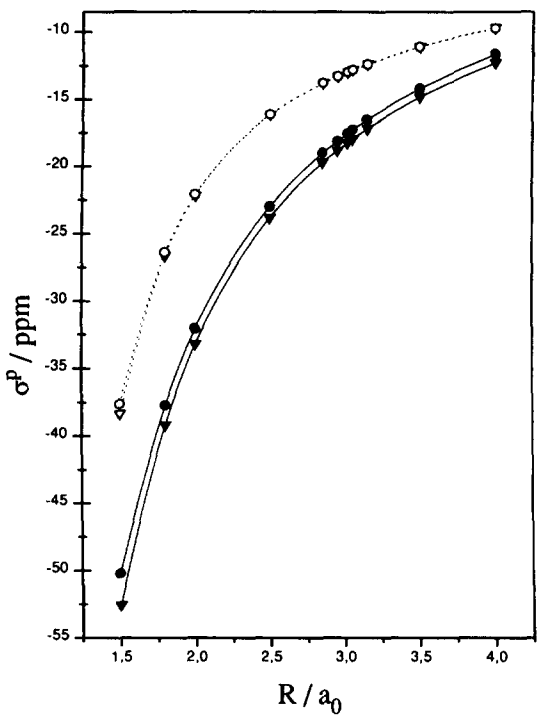


Fig. 1. LiH: The paramagnetic contributions to the nuclear magnetic shielding constants of Li(— ▼ —) at the RPA and (—●—) at the CCSDPPA level and of H (· · · ▽ · · ·) at the RPA and (· · · ○ · · ·) at the CCSDPPA level.

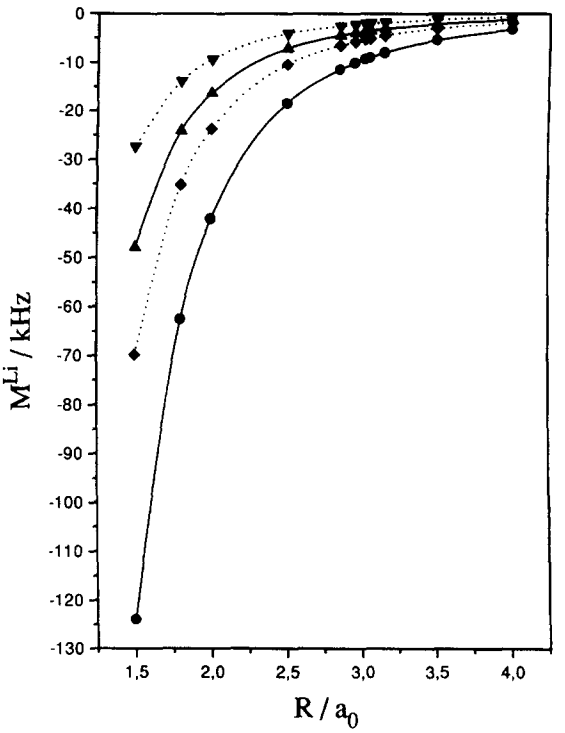


Fig. 2. LiH: Spin–rotation constants M^{Li} of ^{7}LiH (—●—), ^{7}LiD (· · · ◆ · · ·), ^{6}LiH (— ▲ —) and ^{6}LiD (· · · ▼ · · ·) at the CCSDPPA level.

In Fig. 2 the $M^{Li}(R)$ curves (at the CCSDPPA level) for the different isotopomers and in Fig. 3 and the $M^{H/D}(R)$ curves for ^{7}LiH/D are plotted. In general, the nuclear contribution to the spin–rotation constant (Eq. (2)) varies as R^{-3} with the bond length R, whereas the geometry dependence of the electronic term is the product of the R^{-2} dependence arising from the moment of inertia and an a priori unknown geometry dependence of $\sigma^p(R)$. The behaviour of the total spin–rotation constant therefore depends on the form of the geometry dependence of $\sigma^p(R)$ and whether it dominates over the R^{-2} dependence arising from the moment of inertia. In the

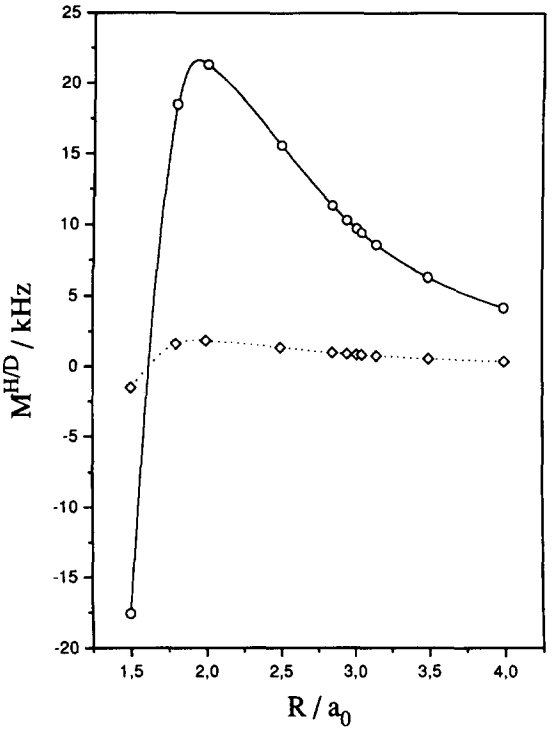


Fig. 3. LiH: Spin–rotation constants $M^{H/D}$ of ^{7}LiH (—○—) and ^{7}LiD (· · · ◇ · · ·) at the CCSDPPA level.

case of LiH, $M^{Li}(R)$ is dominated by the electronic contribution and therefore negative. The $M^{Li}(R)$ curves of the four isotopomers differ in the absolute values but exhibit a similar geometry dependence. $M^{H/D}(R)$ of ^{7}LiH/D is positive for most values of R, but becomes negative for very small bond lengths. Furthermore $M^{H}(R)$ changes much more with the geometry than $M^{D}(R)$. The latter is also almost zero : a result of the near cancellation of the nuclear and electronic terms. The $M^{H/D}(R)$ curves of ^{6}LiH/D are virtually identical to the ones of ^{7}LiH/D since the change in the moment of inertia is very small.

The radial wavefunctions $\chi_{v,J}$ were obtained from Eq. (7) using a cubic taut spline fit [42] to the ab initio potential of Partridge and Langhoff [43]. Averaged values of the spin–rotation and direct spin–spin coupling constants for the six lowest vibrational and the first rotational state of some LiH isotopomers are given in Table 2. Although the nuclear quadrupole coupling study [13] provides data for all vibrational levels supported by the potential energy surface, we limited our analysis to the six lowest vibrational levels, as our calculations are based on a single reference wavefunction, and this is not able to describe the dissociation limit correctly. The absolute values of the averaged spin–rotation constants decrease monotonically with increasing vibrational quantum number v. This is to be expected as the average bond length normally increases with the vibrational quantum number and $M^{Li}(R)$ as well as $M^{H/D}(R)$ (Figs. 2 and 3) approach zero with increasing internuclear distance. The dependence on the rotational quantum numbers, as shown for the ground vibrational state of ^{7}LiH in Table 3, is found to be rather weak.

Table 4 compares the magnetic hyperfine structure constants for the isotopomers of LiH in the first two rotational and vibrational states calculated at the CCSDPPA level with the experimental values. The latter were obtained by Rothstein [27] and improved (for ^{7}LiH and ^{7}LiD) by Freeman et al. [28] by using a fit of molecular beam high resolution electric

Table 2
Vibrational dependence of the spin–rotation constants, M^{Li} and $M^{H/D}$, and the direct spin–spin coupling constants, D, for some LiH isotopomers in the $J = 1$ rotational state calculated at the CCSDPPA level (all values in kHz) [a]

v	^{7}LiH			^{7}LiD			^{6}LiH			^{6}LiD		
	M^{Li}	M^{H}	D	M^{Li}	M^{D}	D	M^{Li}	M^{H}	D	M^{Li}	M^{D}	D
0	−9.14	9.45	11.27	−5.16	0.82	1.73	−3.53	9.64	4.26	−2.03	0.85	0.65
1	−8.95	9.09	10.96	−5.09	0.80	1.70	−3.46	9.27	4.14	−2.00	0.83	0.64
2	−8.75	8.72	10.64	−5.00	0.77	1.67	−3.38	8.89	4.03	−1.96	0.80	0.63
3	−8.51	8.35	10.33	−4.91	0.75	1.63	−3.29	8.51	3.91	−1.93	0.78	0.62
4	−8.25	7.98	10.03	−4.81	0.72	1.59	−3.18	8.13	3.79	−1.89	0.75	0.60
5	−7.93	7.60	9.75	−4.70	0.70	1.56	−3.06	7.74	3.68	−1.84	0.73	0.59

[a] Values for other isotopic species or rovibrational states can be obtained from the authors.

Table 3
Rotational dependence of M^{Li}, M^{H} and D of ^{7}LiH in the ground vibrational state calculated at the CCSDPPA level (all values in kHz) [a]

J	M^{Li}	M^{H}	D
1	−9.14	9.45	11.27
2	−9.13	9.43	11.25
3	−9.10	9.42	11.23
4	−9.07	9.39	11.20
5	−9.03	9.36	11.16
10	−8.74	9.13	10.86
15	−8.28	8.76	10.40
20	−7.70	8.28	9.80
25	−7.03	7.71	9.10
30	−6.31	7.08	8.33
α [b]	−9.15	9.45	11.28
$\beta_1/10^{-3}$ [b]	3.728	−2.930	−3.861
$\beta_2/10^{-7}$ [b]	−7.392	3.958	7.915

[a] See footnote a of Table 2.
[b] Coefficients in the fit function for the J dependence: $\alpha + \beta_1 J(J+1) + \beta_2 J^2(J+1)^2$.

resonance spectra to Ramsey's hyperfine structure Hamiltonian [44]. For completeness we have also included the previously calculated nuclear quadrupole coupling constants [10]. Agreement is quite satisfactory. The calculated values of $M^{H/D}_{v,J}$ for all isotopic species as well as the values of $M^{Li}_{v,J}$ in ^{7}LiD, ^{6}LiH and ^{6}LiD are within or close to the experimental error bars. Although the values of $M^{Li}_{v,J}$ in ^{7}LiH lie about 1 kHz below the experimental ones, their vibrational dependence is in fair agreement with experiment.

The deuteron and ^{6}Li quadrupole coupling constants calculated in Ref. [10] were in very good agreement with the experimental values, however the results for the ^{7}Li nucleus deviated uniformly from experiment. If the value for quadrupole moment of ^{7}Li nucleus from Ref. [45] is used instead of the one in Ref. [10], perfect agreement with experiment [28] is obtained for the ^{7}Li quadrupole coupling constants (given in brackets in Table 4).

3.2. HF

The electronic part of the spin–rotation constant is proportional to the paramagnetic term of the nuclear magnetic shielding constant σ^p (Eq. (2)) and

Table 4
Comparison of the calculated magnetic hyperfine constants (M^{Li}, $M^{H/D}$ and D) and the nuclear quadrupole coupling constants $eQ_{Li/D}q^{Li/D}_{v,J}/h$ for LiH with experimental values (all values in kHz) [a,b]

	v,J		$eQ_{Li}q^{Li}_{v,J}/h$ [c]	$eQ_{D}q^{D}_{v,J}/h$ [c]	M^{Li}	$M^{H/D}$	D
^{7}LiH	0,1	calc.	329.0(346.8)		−9.143	9.446	11.27
		exp. [d]	346.780 ± 0.036		−10.054 ± 0.007	9.345 ± 0.027	11.346 ± 0.007
	0,2	calc.	329.0(346.8)		−9.128	9.434	11.25
		exp. [d]	346.64 ± 0.5		−10.034 ± 0.003	9.31 ± 0.02	11.329 ± 0.012
	1,1	calc.	313.0(330.0)		−8.956	9.087	10.96
		exp. [d]	330.8 ± 0.4		−9.88 ± 0.04	8.87 ± 0.20	11.03 ± 0.08
^{7}LiD	0,1	calc.	331.1(349.0)	33.5	−5.164	0.821	1.73
		exp. [d]	348.942 ± 0.037	34.213 ± 0.033	−5.682 ± 0.005	0.819 ± 0.006	1.743 ± 0.007
	1,1	calc.	318.9(336.1)	33.2	−5.086	0.798	1.70
^{6}LiH	0,1	calc.	6.7		−3.534	9.642	4.26
		exp. [e]	7.2 ± 0.8		−3.9 ± 0.5	9.3 ± 0.7	
	1,1	calc.	6.4		−3.461	9.271	4.14
		exp. [e]	7.2 ± 0.8		−3.9 ± 0.5	9.3 ± 0.7	
^{6}LiD	0,1	calc.	6.8	33.5	−2.028	0.851	0.65
		exp. [e]	7.5 ± 1.0	33 ± 1	−3.2 ± 1.0	1.3 ± 0.5	
	1,1	calc.	6.5	33.2	−1.996	0.827	0.64
		exp. [e]	7.5 ± 1.0	33 ± 1	−3.2 ± 1.0	1.3 ± 0.5	

[a] See footnote a of Table 2.
[b] The sign convention for M from Ref. [18] is used. The sign of the experimental values are altered accordingly.
[c] The quadrupole coupling constants are taken from Ref. [10]. The values in parentheses are obtained by using the nuclear electric quadrupole moment ($Q_{Li} = -4.26 \times 10^{-30}$ m^2) from Ref. [45].
[d] Ref. [28].
[e] Ref. [27].

the total nuclear magnetic shielding constant σ for the equilibrium bond length can thus be derived from a measured spin–rotation constant and a calculated diamagnetic contribution to σ, if the vibrational dependence is properly taken into account. In 1968 Hindermann and Cornwell [46] obtained equilibrium nuclear magnetic shielding values for F and H in the HF molecule from spin–rotation constants measured in the ground vibrational and first excited rotational state of HF. Their value, $\sigma_F = 419.7 \pm 6$ ppm, is frequently used as a standard in the comparison of results of various theoretical approaches. Moreover the experimental absolute scale of the nuclear magnetic shielding constants σ of F is based on it. Unlike the molecular systems studied in the following section, the correlation contribution to the magnetic properties of HF are relatively small (see e.g. Refs. [16,22]) and in general calculations approach the experimental data at the CHF level. Results of numerous calculations for the nuclear magnetic shielding constant of F in HF lie, therefore, within the experimental error bars of 6 ppm set in Ref. [46], and, if some care is taken in choosing a basis set, even the gauge dependence problem can be minimized (see e.g. Refs. [41,47]). On the other hand, calculations of spin–rotation constants [48,49,16], seem to give poor agreement with experiment (see Table 8). The difference in the percentage errors between the total nuclear magnetic shielding and the total spin–rotation constant arise because both are the sum of two terms (Eq. (2)). For example an error of ~ 6 ppm in σ_F^p, which is 1.5% of the total F magnetic shielding constant, leads to an error of ~ 30 kHz or 10% in the spin–rotation constant of F. Furthermore the 6 ppm error bars of Hindermann and Cornwell [46] come mainly from the uncertainty in the theoretically derived diamagnetic contribution and rovibrational corrections and not from the experiments. The error bars on the recent measurements of M^F and M^H [29,30,3], on the other hand, are much smaller. The most direct and precise way of assessing the quality of methods for the prediction of nuclear magnetic shielding constants is therefore to calculate the spin–rotation constants for the actually measured rovibrational states rather than comparing with not directly measurable nuclear magnetic shielding constants. In particular the large increase in M^F from the $v = 0$ to the $v = 1$ vibrational state is a challenge for theory.

In our previous studies of magnetic properties [21,22] basis set were presented, which gave converged results for the magnetizability and nuclear magnetic shielding constants of several molecules at the correlated SOPPA level. These basis sets were labelled $sp_t d$ for hydrogen and $sp_{tt}d_{td}f_2$ for the other atoms. For the present study the basis set for fluorine was augmented by one set of diffuse p and diffuse d functions with exponents forming a geometric series with two preceding functions ($\xi_p = 0.063$, $\xi_d = 0.068$) and the exponents of the two sets of f functions were modified ($\xi_{f1} = 1.8$, $\xi_{f2} = 0.6$). In addition the set of tight p functions ($\xi_p = 6.269$) on H was removed, since this did not alter the results. The final basis consists of 102 Gaussian basis functions. The results for σ_H^p, σ_F^p, M^F and $M^{H/D}$ calculated at the equilibrium distance, $R = 1.7333\, a_0$ are given in Table 5. Electron correlation contributions are small for both σ_H^p and σ_F^p and lead to a reduction of the absolute values; the correlation correction to σ_F^p at CCSDPPA level, e.g., is $\sim 4\%$ similar to the one for σ_{Li}^p in LiH.

The calculations were done for 12 internuclear distances R in the range between 1.0 a_0 and 3.0 a_0. This is sufficient for the lowest vibrational levels. At

Table 5
Dependence of σ^p (in ppm) and M (in kHz) of HF on the level of the polarization propagator approximation for the equilibrium bond length ($R_e = 1.7333\, a_0$)

			HF		DF	
	σ_F^p	σ_H^p	M^F	M^H	M^F	M^D
RPA	−68.32	−80.07	−313.16	69.30	−164.59	5.59
SOPPA	−66.34	−79.06	−302.48	75.08	−158.98	6.06
CCDPPA	−66.77	−79.15	−304.80	74.57	−160.20	6.02
CCSDPPA	−65.45	−79.10	−297.68	74.86	−156.45	6.04

larger internuclear distances methods based on a single determinant wavefunction become unreliable and a multiconfigurational version of the PPA should be used. The $\sigma_F^p(R)$ and $\sigma_H^p(R)$ curves at the RPA and CCSDPPA level are shown in Fig. 4. Different from LiH the RPA and CCSDPPA curves for $\sigma_F^p(R)$ are not parallel and the rovibrational corrections to M^F will therefore depend on the level of approximation. From $\sigma_F^p(R)$ and $\sigma_H^p(R)$ the spin–rotation functions $M^F(R)$ and $M^{H/D}(R)$ were calculated from Eq. (2). In Figs. 5 and 6 the $M^F(R)$ and $M^{H/D}(R)$ curves for the two isotopomers at the CCSDPPA level are shown. $M^{H/D}(R)$ is positive and falls off monotonically to zero. Again, as in the case of LiH, $M^H(R)$ is larger than $M^D(R)$ and varies more with the internuclear distance. Similar to LiH, $M^F(R)$ is also dominated by the electronic contribution. Contrary to LiH, however, $\sigma_F^p(R)$ does not approach zero within the investigated range of internuclear distances, but approaches a pole. It dominates over the R^{-2} dependence arising from the moment of inertia and the calculated total spin–rotation constant $M^F(R)$ thus also falls of monotonically towards $-\infty$ in the investigated range of internuclear distances.

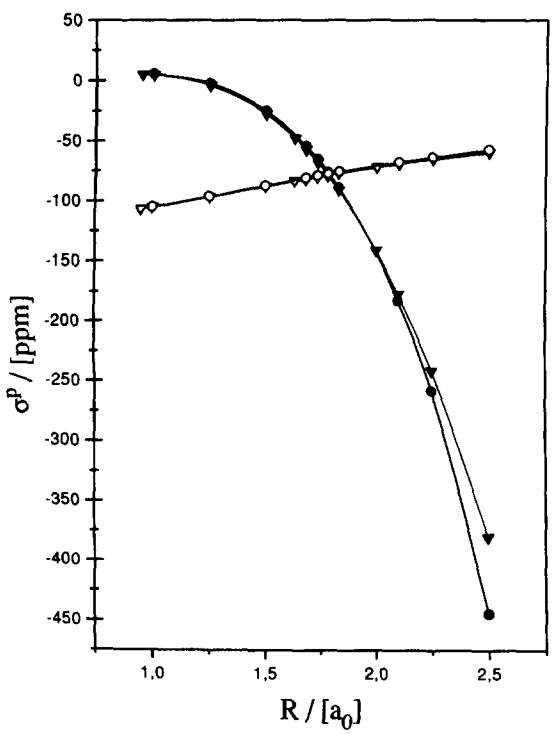


Fig. 4. HF: The paramagnetic contributions to the nuclear magnetic shielding constants of F (—▼—) at the RPA and (—●—) at the CCSDPPA level and of H (—∇—) at the RPA and (—○—) at the CCSDPPA level.

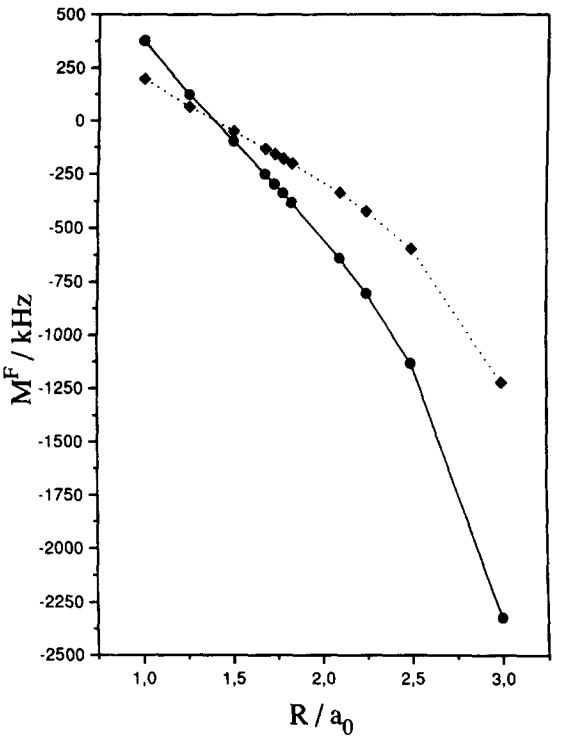


Fig. 5. HF: Spin–rotation constants M^F of HF (—●—) and DF (· · · ◆ · · ·) at the CCSDPPA level.

Rovibrational averaging (Eq. (6)) of the spin–rotation constants and the direct spin–spin interaction constants, calculated from Eq. (5), was done for the six lowest vibrational levels of ^{19}FH and ^{19}FD. The radial wavefunctions were obtained as solutions of the rovibrational Schrödinger equation (7) using the RKR potential curve of Di Lonardo and Douglas [50]. The dependence on the vibrational quantum number v of the magnetic hyperfine structure constants M^F, $M^{H/D}$ and D of ^{19}FH and ^{19}FD in the $J = 1$ rotational state is shown in Table 6 for the RPA and CCSDPPA levels. In Table 7 the weak rotational dependence of M^F is demonstrated for the $v = 0$ and $v = 1$ vibrational states. Vibrational averaging increases the absolute value of the spin–rotation constant of fluorine contrary to what we have found for lithium in LiH, but decreases the absolute

Table 6
Vibrational dependence of the spin–rotation constants, M^F and $M^{H/D}$, and the direct spin–spin interaction constants, D in $H^{19}F$ and $D^{19}F$ in the $J = 1$ rotational state (all values in kHz) [a]

v	HF					DF				
	M^F		M^H		D	M^F		M^H		D
	RPA	CCSDPPA	RPA	CCSDPPA		RPA	CCSDPPA	RPA	CCSDPPA	
0	−337.95	−325.61	68.64	74.25	143.58	−173.92	−166.93	5.56	6.01	22.17
1	−389.04	−384.77	67.17	72.80	137.74	−192.95	−188.80	5.47	5.93	21.53
2	−442.97	−449.89	65.49	71.16	131.83	−213.22	−212.49	5.37	5.84	20.86
3	−498.40	−521.80	63.85	69.47	126.23	−233.54	−237.41	5.29	5.75	20.23
4	−557.73	−604.19	61.85	67.44	120.43	−254.63	−264.59	5.19	5.65	19.60
5	−619.86	−696.81	59.68	65.37	114.73	−276.76	−294.56	5.08	5.54	18.96

[a] See footnote a of Table 2.

value of the spin–rotation constant of the hydrogen isotopomers and of the direct spin–spin coupling constants. This is again a consequence of the increase in the averaged bond length and the dependence of $M^F(R)$ and $M^{H/D}(R)$ on R as shown in Figs. 5 and 6. As mentioned above, the RPA and CCSDPPA σ_F^p curves are not parallel; the rovibrational corrections to M^F are, e.g., larger at the CCSDPPA level than at the RPA level. Furthermore, the difference of M^F for two vibrational states increases much faster at the CCSDPPA level.

In Table 8, our calculated spin–rotation constant are compared with experimental data and the results of earlier calculations. For the ^{19}FH isotopomer, the difference between the CCSDPPA values of the spin–rotation constants calculated at the equilibrium bond length and those derived from experiments is 17.91 kHz for M_{eq}^F and 3.15 kHz for M_{eq}^H which may be compared with 32.66 kHz for M_{eq}^F and 2.72 kHz for M_{eq}^H at the RPA level. The CCSDPPA correlation contribution improves thus the agreement with the experimental value for M_{eq}^F. Also the rovibrational correction for the $v = 0$, $J = 1$ state, calculated from the CCSDPPA curve, is in good agreement with the value derived from experiment. Comparing the $M_{1,1}^F - M_{0,1}^F$ differences, shows that the large increase in M^F is reproduced fairly well at both levels of theory, although the RPA result (−51.09 kHz) is in slightly better agreement with experiment (−53.33 kHz) than the CCSDPPA value (−59.16 kHz). One might say that at the region close to the equilibrium internuclear distance the CCSDPPA results are superior to RPA values, while at the larger distances the CCSDPPA $M^F(R)$ curve is too steep. This can also be seen from the derivatives of $M^F(R)$ with respect to $\xi = (R - R_e)/R_e$. At $R = 1.7333\ a_0$ the first derivatives of $M^F(R)$ are -1520 ± 1 for the CCSDPPA curve and -1460 ± 1 at the RPA level and the second derivatives are -300 ± 10 (CCSDPPA) and $+35 \pm 2$ (RPA). These values should be compared with those derived from experiment [3] [1]: -1450 ± 50 and -400 ± 200, respectively. For the sake of

Table 7
Rotational dependence of M^F of HF in the ground and first excited vibrational state calculated at the RPA and CCSDPPA levels (all values in kHz) [a]

J	$v = 0$		$v = 1$	
	RPA	CCSDPPA	RPA	CCSDPPA
1	−337.95	−325.61	−389.04	−384.77
2	−338.56	−326.26	−389.69	−385.48
3	−339.48	−327.23	−390.65	−386.54
4	−340.70	−328.53	−391.94	−387.96
5	−342.23	−330.16	−393.55	−389.74
10	−354.53	−343.23	−406.51	−404.10
15	−374.68	−364.79	−427.81	−427.90
20	−402.95	−395.32	−457.76	−461.88
α [b]	−337.64	−325.28	−388.72	−384.42
$\beta_1/10^{-2}$ [b]	−15.279	−16.184	−16.079	−17.767
$\beta_2/10^{-6}$ [b]	−6.4268	−11.684	−8.5437	−12.437

[a] See footnote a of Table 2.
[b] Coefficients in the fit function for the J dependence: $\alpha + \beta_1 J(J+1) + \beta_2 J^2(J+1)^2$.

[1] Note, that there are opposite signs in the definition of our M^K values and the corresponding C^K values in Ref. [3].

completeness we calculated also at the equilibrium distance the first derivatives of $M^{H}(R)$ with respect to ξ (-343 ± 1 at the CCSDPPA level, -344 ± 1 at the RPA level) and the second derivatives (1027 $\pm$ 5 CCSDPPA, 1014 $\pm$ 5 RPA). The corresponding experimental values [3] are -260 ± 20 and 500 ± 200. The rotational dependence of M^{F} and M^{H} in the $v = 1$ state can be compared with the experimental data of Bréant [51] measured by saturated absorption spectroscopy. Fitting the experimental data for the $J = 1$–5 states to the function $\alpha v + \beta J(J+1)$, Bréant obtained $\beta = 0.17 \pm 0.05$ kHz for M^{F} and $\beta = 0.036 \pm 0.030$ kHz for M^{H} in good agreement with our calculated results : $\beta = 0.161$ kHz (RPA) and $\beta = 0.178$ kHz (CCSDPPA) for M^{F} and $\beta = 0.0345$ kHz (RPA) and $\beta = 0.0343$ kHz (CCSDPPA) for M^{H}.

The relative size of the magnetic and electric hyperfine constants in the two lowest vibrational states is well known from experiment [3,29]. The ^{19}F nucleus does not possess a quadrupole moment, and the hyperfine structure of ^{19}FH is therefore determined by magnetic interactions alone. From the smooth dependence on the vibrational quantum number of all magnetic hyperfine structure constants, we can expect, that the hyperfine structure pattern will also exist in higher rovibrational levels. For ^{19}FD, the dominant interactions causing the hyperfine splitting in its rovibrational spectrum are the deuteron quadrupole coupling (the deuteron quadrupole coupling constant for the $v = 0$, $J = 1$ state is 354.238 kHz [29]) and the spin–rotation interaction of F. The strength of the spin–rotation interaction of F increases markedly with v. Deuteron quadrupole cou-

Table 8
Comparison of the calculated spin–rotation constants of HF and DF with experimental values and results of other calculations (all values in kHz) [a,b]

		RPA	CCSDPPA	exp.
HF	M^{F}_{eq}	−313.15 [c]	−297.68 [d]	−280.49 ± 0.04 [e]
	$M^{F}_{0,1}$	−337.95	−325.61	−307.65 ± 0.02 [e]
	$M^{F}_{eq} - M^{F}_{0,1}$	24.80 [f]	27.93	27.16 ± 0.06 [e]
	$M^{F}_{0,2}$	−338.56	−326.26	−308.263 ± 0.019 [g]
	$M^{F}_{1,1}$	−389.04	−384.77	−361.20 ± 0.02 [e]
	M^{H}_{eq}	69.30 [h]	74.86 [i]	72.02 ± 0.04 [e]
	$M^{H}_{0,1}$	68.64	74.25	71.10 ± 0.02 [e]
	$M^{H}_{eq} - M^{H}_{0,1}$	0.66 [j]	0.61	0.92 ± 0.06 [e]
	$M^{H}_{0,2}$	68.50	74.11	71.033 ± 0.027 [g]
	$M^{H}_{1,1}$	67.17	72.80	69.33 ± 0.03 [e]
DF	M^{F}_{eq}	−164.58	−156.45	
	$M^{F}_{0,1}$	−173.90	−166.93	−158.356 ± 0.001 [k]
	M^{D}_{eq}	5.59	6.03	
	$M^{D}_{0,1}$	5.56	6.01	5.755 ± 0.019 [k]

[a] See footnote a of Table 2.

[b] The sign conversion for M from Ref. [18] is used. The sign of the experimental values are altered accordingly.

[c] Cf. the CHF result (−312.4 kHz) of Stevens and Lipscomb [48]; the GIAO-CHF result (−294.9 kHz) of Ditchfield [49]; the CHF result (−306.99 kHz obtained from σ^{p} by use of Eq. (2)) of Höller and Lischka [47]; the RPA result (−313.33 kHz) of Oddershede and Geertsen [16].

[d] Cf. the SOPPA result (−304.36 kHz) of Oddershede and Geertsen [16].

[e] Ref. [3].

[f] Cf. the CHF result (23.6 kHz) of Stevens and Lipscomb [48].

[g] Ref. [30].

[h] Cf. the CHF result (70.65 kHz) of Stevens and Lipscomb [48]; the GIAO-CHF result (90.2 kHz) of Ditchfield [49]; the CHF result (74.30 kHz obtained from σ^{p} by use of Eq. (2)) of Höller and Lischka [47]; the RPA result (81.50 kHz) of Oddershede and Geertsen [16].

[i] Cf. the SOPPA result (90.76 kHz) of Oddershede and Geertsen [16].

[j] Cf. the CHF result (0.68 kHz) of Stevens and Lipscomb [48].

[k] Ref. [29].

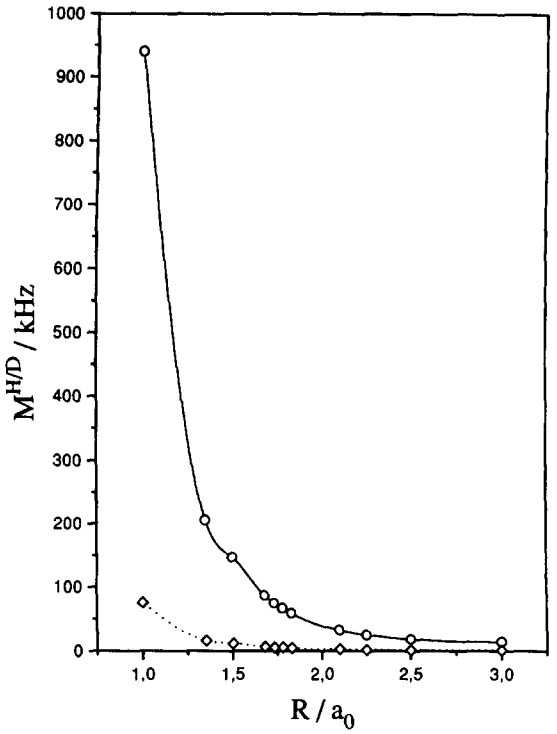


Fig. 6. HF: Spin–rotation constants $M^{H/D}$ of HF (—○—) and DF (· · · ◇ · · ·) at the CCSDPPA level.

pling constants for higher rovibrational states are therefore necessary for the determination of the hyperfine splitting in these states.

After comparing our calculated results directly with the experimental data, we, nevertheless, should discuss our values for the nuclear magnetic shielding constants of HF. In the absence of sufficient experimental data, Hindermann and Cornwell [46] had to estimate the rovibrational correction to the spin–rotation constant from an earlier CHF calculation [48] for three different bond lengths. The recent measurements [3] in an excited vibrational state allowed Bass et al. [3] to obtain with high accuracy an extrapolated value for the equilibrium geometry from experimental data alone. Their values for M_{eq}^{F} and for M_{eq}^{H} lead to newer experimental values for the paramagnetic contribution to the nuclear magnetic shielding constants of HF : $\sigma_F^p = -62.26 \pm 0.01$ ppm and $\sigma_H^p = -79.59 \pm 0.01$ ppm. These can be compared with our RPA ($\sigma_F^p = -68.32$ ppm, $\sigma_H^p = -80.07$ ppm) and CCSDPPA results ($\sigma_F^p = -65.45$ ppm, $\sigma_H^p = -79.10$ ppm). In order to arrive at something like an experimental value of the total nuclear magnetic shielding constant, one has to combine the experimental paramagnetic contribution with a theoretically derived diamagnetic term. The diamagnetic term is in general rather insensitive to correlation contributions and basis set effects. Using a sum-over-states expression for the diamagnetic term [52] we calculated, at the CCSDPPA level, $\sigma_F^d = 480.29$ ppm and $\sigma_H^d = 108.22$ ppm with an estimated remaining basis set error and correlation contribution of ± 2 ppm. Added to the experimental paramagnetic term we thus obtain $\sigma_F = 418.0 \pm 2.0$ and $\sigma_H = 28.6 \pm 2.0$ as semi-experimental nuclear shielding constants of HF at the equilibrium internuclear distance.

3.3. CH^+ and BH

The isoelectronic molecules CH^+ and BH are known to have magnetic properties which are unusual for closed shell diatomic molecules: paramagnetism, antishielding and extremely large spin–rotation constants of the heavier nucleus. The magnetic properties of the BH molecule have for many years been the subject of theoretical investigations. The first theoretical prediction of their anomalous character was made by Stevens and Lipscomb in 1965 [53] and confirmed one year later by Hegstrom and Lipscomb [54]. Recently, growing interest in the paramagnetism of closed shell diatomic hydrides has lead to new theoretical investigations of the magnetic properties of these molecules (see e.g. Refs. [33,55,56]). Fowler and Steiner [33] pointed out that the spin–rotation constant of C in $^{13}CH^+$ is even more negative than the extreme value of M^B in ^{11}BH. Using the CHF method they obtained, at the equilibrium nuclear configuration, $M^C = -1452$ kHz for $^{13}CH^+$ and $M^B = -485$ kHz for ^{11}BH. In the present study we investigated the effects of electronic correlation and rovibrational averaging on the spin–rotation constants of these molecules. Moreover, we provide hyperfine structure parameters complementary to the nuclear electric quadrupole coupling data of CH^+ [13] and BH [11].

Fowler and Steiner employed in their CHF study Sadlej's polarized basis sets [57] for the C, B and H atoms. These basis sets developed for electric properties are less appropriate for the calculation of mag-

Table 9
Dependence of σ^{p} (in ppm), M (in kHz) and the rovibrational correction to M (in kHz) of $^{13}CH^{+}$ and ^{11}BH on the level of the polarization propagator approximation for the equilibrium bond lengths ($R_e(^{13}CH^{+}) = 2.137\,a_0$, $R_e(^{11}BH) = 2.333\,a_0$)

	$^{13}CH^{+}$						^{11}BH					
	$\sigma_C^p(R_e)$	$M^C(R_e)$	$M_{0,1}^C - M^C(R_e)$	$\sigma_H^p(R_e)$	$M^H(R_e)$	$M_{0,1}^H - M^H(R_e)$	$\sigma_B^p(R_e)$	$M^B(R_e)$	$M_{0,1}^B - M^B(R_e)$	$\sigma_H^p(R_e)$	$M^H(R_e)$	$M_{0,1}^H - M^H(R_e)$
RPA	−1541.60	−1486.95	7.35	−27.46	86.28	−0.77	−475.99	−492.59	7.46	−33.19	15.90	0.05
SOPPA	−1593.94	−1537.71	7.21	−28.10	83.82	−0.73	−522.58	−541.59	7.56	−34.25	12.43	0.12
CCDPPA	−1259.37	−1213.23	9.95	−30.29	75.37	−0.75	−451.47	−466.80	7.10	−33.61	14.53	0.10
CCSDPPA	−1241.93	−1196.34	10.19	−30.71	73.75	−0.72	−445.64	−460.67	6.92	−33.90	13.58	0.12

netic properties, even so, complete uncontraction reduces the dependence of the results on the gauge origin. Here, we used the $sp_{tt}d_{td}f_2$ basis set for the C and B atom from our previous work on magnetic properties [21,22] and the same basis set for hydrogen as in HF. The values of the paramagnetic contribution to the nuclear magnetic shielding constants and of the spin–rotation constants of $^{13}CH^+$ and ^{11}BH calculated at the equilibrium bond lengths are given in Table 9 as a function of the level of approximation. Electron correlation increases the paramagnetic contribution to the nuclear shielding constant of hydrogen in both molecules. The effect is more pronounced for CH^+, where the changes are up to ~ 12% at the CCSDPPA level, than for BH, where it amounts to only 2–3% and the largest correlation contribution is obtained at the SOPPA level. For the paramagnetic term of the nuclear shielding of C and B one finds the same dependence on the level of the PPA as was previously observed for the paramagnetic contribution to the magnetizability [55]. The SOPPA results for these near-degenerate systems are affected by too small excitation energies while the CCDPPA/CCSDPPA results are found to be satisfactory. The $\sigma_H^p(R)$ and $\sigma_{C/B}^p(R)$ curves were calculated for a range of internuclear distances between 1.13 a_0 and 4.0 a_0 in the case of CH^+ and between 1.3 a_0 and 4.0 a_0 in the case of BH. $\sigma_C^p(R)$ and $\sigma_B^p(R)$ are displayed in Figs. 7 and 8. It is interesting to note that for both molecules $\sigma^p(R)$ goes through a maximum, although its location is different in the two molecules. In the case of BH the maximum appears at an internuclear distance larger than the equilibrium value whereas in CH^+ it lies towards smaller bond lengths. It can also be seen that the correlation effects are slightly distance dependent.

In Figs. 9 and 11 the $M^C(R)$ and $M^B(R)$ curves (at the CCSDPPA level) for the different isotopomers of CH^+ and BH are plotted. The nuclear g-factor of ^{11}C is negative, which is the origin of the different sign of M^C in $^{11}CH/D^+$ and $^{13}CH/D^+$.

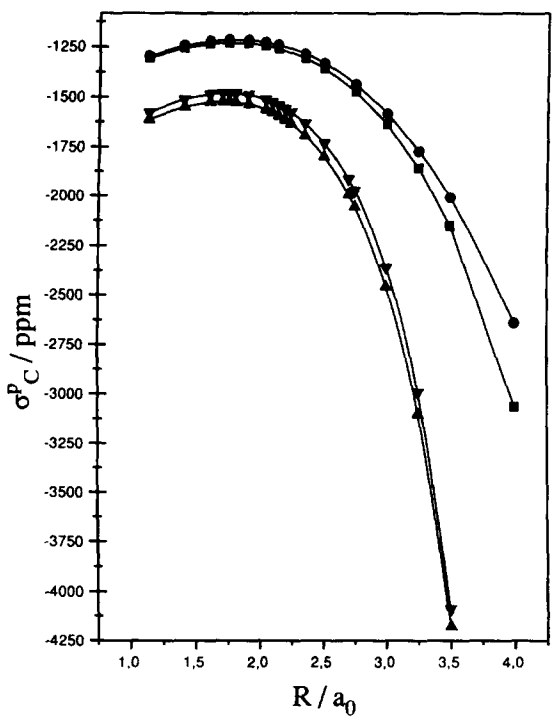


Fig. 7. CH^+: The paramagnetic contributions to the nuclear magnetic shielding constants of C at the RPA (—▼—), the SOPPA (—▲—), the CCDPPA (—■—) and the CCSDPPA (—●—) level.

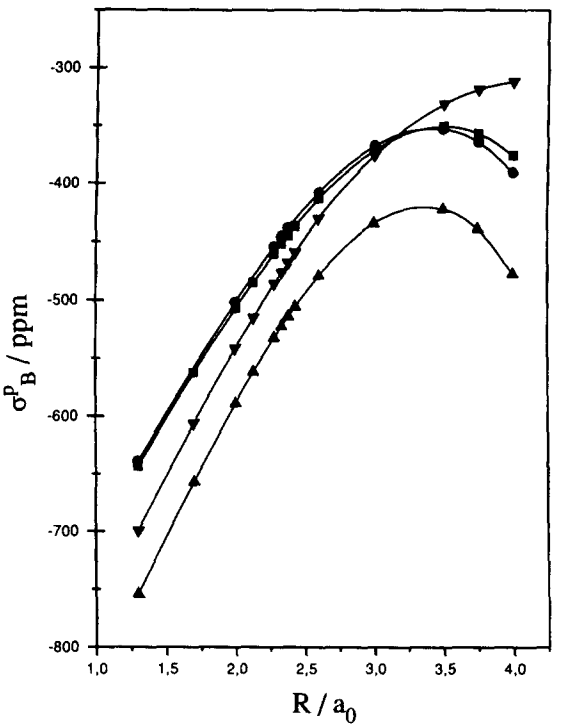


Fig. 8. BH: The paramagnetic contributions to the nuclear magnetic shielding constants of B at the RPA (—▼—), the SOPPA (—▲—), the CCDPPA (—■—) and the CCSDPPA (—●—) level.

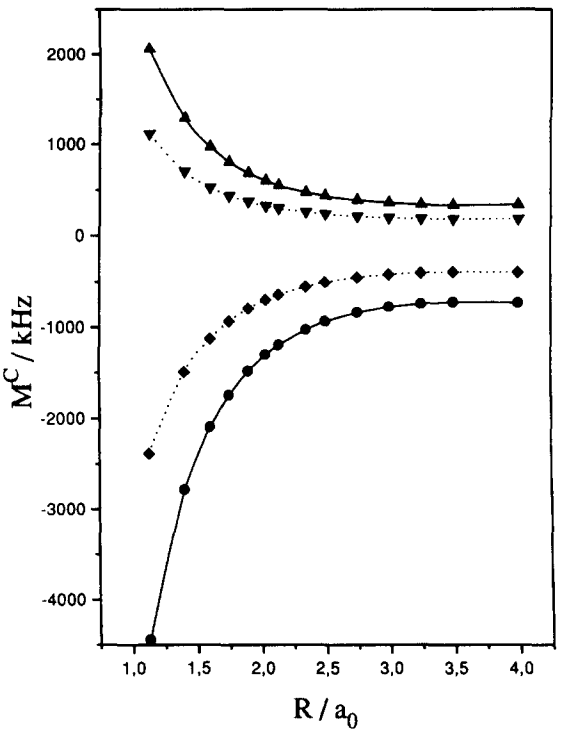


Fig. 9. CH^+: Spin–rotation constants M^C of $^{13}CH^+$ (—●—), $^{13}CD^+$ (· · · ◆ · · ·), $^{11}CH^+$ (—▲—) and $^{11}CD^+$ (· · · ▼ · · ·) at the CCSDPPA level.

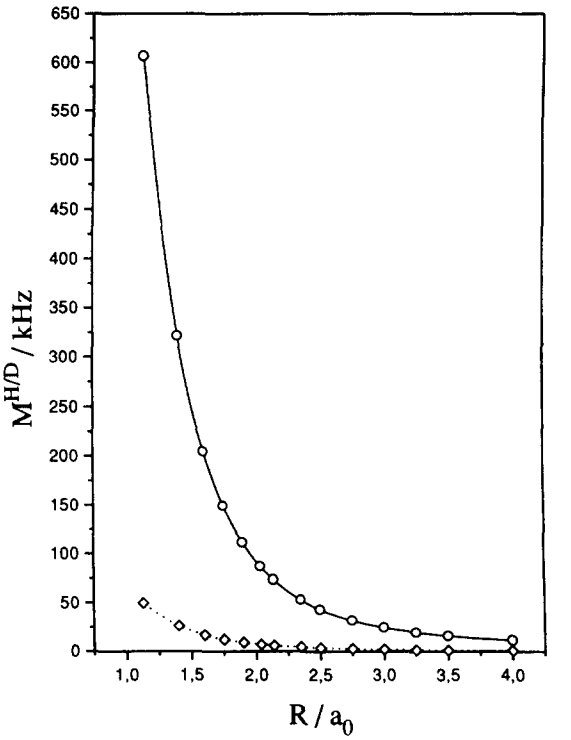


Fig. 10. CH: Spin–rotation constants $M^{H/D}$ of $^{13}CH^+$ (—○—) and $^{13}CD^+$ (· · · ◇ · · ·) at the CCSDPPA level.

For both isotopomers M^C seems to approach zero with increasing internuclear distances, but then goes through a maximum (minimum in the case of $^{11}CH/D^+$) for larger internuclear distances. Contrary to HF, for which $\sigma_F^p(R)$ showed a similar dependence on the bond length, the R^{-2} term coming from the moment of inertia thus determines the form of M^C around the equilibrium internuclear distance. In BH no maximum of M^B is predicted within the investigated range of geometries and M^B thus approaches zero with increasing bond length. Figs. 10 and 12 show that the $M^{H/D}(R)$ curves for $^{13}CH/D^+$ and ^{11}BH are very similar to the ones obtained for HF. $M^{H/D}$ is dominated by the nuclear contribution, which is positive and approaches zero for large internuclear distances.

Table 10
Vibrational dependence of the spin–rotation constants, M^C and $M^{H/D}$, and the direct spin–spin coupling constants, D, for some CH^+ isotopomers in the $J = 1$ rotational state calculated at the CCSDPPA level (all values in kHz) [a]

v	$^{13}CH^+$			$^{13}CD^+$			$^{11}CH^+$			$^{11}CD^+$		
	M^C	M^H	D	M^C	M^D	D	M^C	M^H	D	M^C	M^D	D
0	−1186.15	73.02	20.59	−638.18	6.039	3.178	549.65	73.97	−9.420	299.00	6.185	−1.454
1	−1160.63	70.74	19.82	−627.92	5.903	3.091	537.57	71.64	−9.062	294.14	6.043	−1.413
2	−1135.16	68.38	19.03	−617.86	5.762	3.003	525.88	69.23	−8.703	289.37	5.897	−1.373
3	−1111.12	65.99	18.26	−608.13	5.620	2.916	514.67	66.79	−8.345	284.76	5.750	−1.332
4	−1087.94	63.54	17.48	−598.63	5.476	2.828	503.87	64.29	−7.987	280.26	5.600	−1.292
5	−1065.62	61.03	16.70	−589.40	5.329	2.741	493.47	61.73	−7.627	275.89	5.448	−1.251

[a] See footnote a of Table 2.

The rovibrational wavefunctions were obtained as solutions of Eq. (7), using the accurate potential energy curves of Saxon et al. [58] for the isotopomers of CH^+, and using the hybrid potential of Luh and Stwalley [59] for the isotopomers of BH. The rovibrational averaged spin–rotation constants were then calculated from Eq. (6). The rovibrational corrections to the spin–rotation constants of $^{13}CH^+$ and ^{11}BH in the $v = 0$, $J = 1$ state are given in Table 9 for different levels of the PPA. Rovibrational averaging reduces the absolute values of the spin–rotation constants, as expected from Figs. 9–12, with the exception of M^H in ^{11}BH. Although the rovibrational corrections depend on the level of the PPA, they seem to be converged at the CCSDPPA level. Therefore, the $M^K(R)$ curves based on the CCSDPPA calculations of $\sigma^p(R)$ were used in the calculation (Eq. (6)) of the spin–rotation constants for the other rovibrational states. The results of the rovibrational averaging for some isotopomers of CH^+ and BH are presented in Table 10, Table 11, Tables 13 and 14.

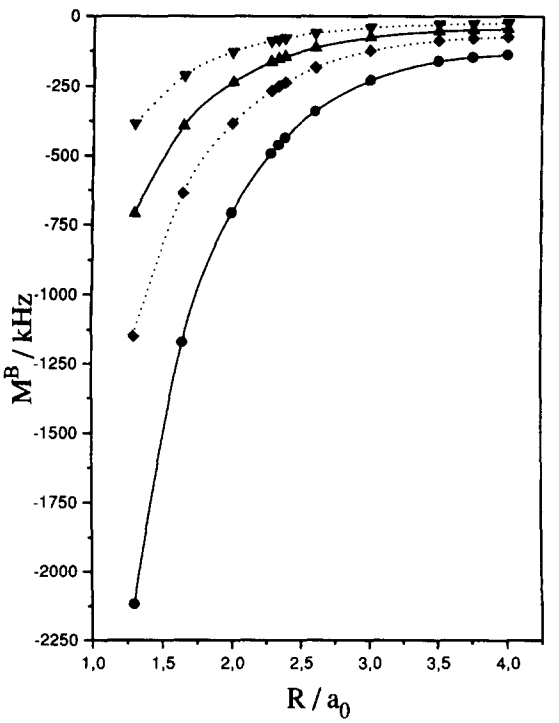


Fig. 11. BH: Spin–rotation constants M^B of ^{11}BH (—●—), ^{11}BD (· · · ◆ · · ·), ^{10}BH (— ▲ —) and ^{10}BD (· · · ▼ · · ·) at the CCSDPPA level.

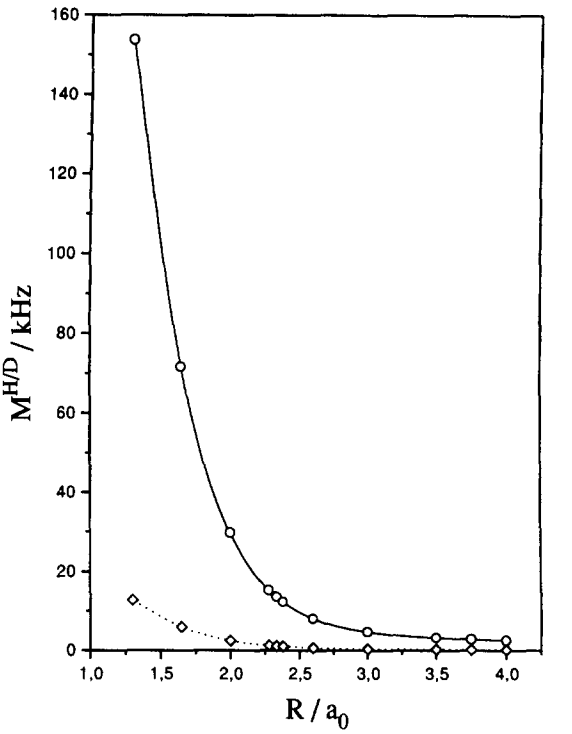


Fig. 12. BH: Spin–rotation constants $M^{H/D}$ of ^{11}BH (—○—) and ^{11}BD (· · · ◇ · · ·) at the CCSDPPA level.

We now turn our attention to the results for the CH^+ molecule. In order to provide the spin–rotation constants which are complementary to the recent quadrupole coupling calculation [13], the spin–rotation constants of $^{11}CH^+$, $^{13}CH^+$ and their deuterium isotopomers were investigated. Amongst the C isotopes active in magnetic interactions, the radioactive ^{11}C nucleus is the only one, which posses a nuclear quadrupole moment. This motivated the hyperfine structure study for the ^{11}C isotopic species. The dependence of $M^C_{v,J}$ and $M^H_{v,J}$ on the vibrational quantum number ($v = 1$–6) is shown in Table 10 for the $J = 1$ rotational state. In all isotopomers the absolute values of the spin–rotation and the direct nuclear spin–spin coupling constants decrease monotonically with increasing vibrational quantum number v. The weak dependence of the magnetic coupling constants on the rotational quantum number J is illustrated in Table 11 for the ground vibrational state.

Table 11
Rotational dependence of M^C, $M^{H/D}$ and D of some CH^+ isotopomers in the ground vibrational state calculated at the CCSDPPA level (all values in kHz) [a]

J	$^{13}CH^+$			$^{13}CD^+$			$^{11}CH^+$			$^{11}CD^+$		
	M^C	M^H	D	M^C	M^D	D	M^C	M^H	D	M^C	M^D	D
1	−1186.15	73.02	20.59	−638.18	6.039	3.178	549.65	73.97	−9.420	299.00	6.185	−1.454
2	−1185.35	72.92	20.57	−637.95	6.034	3.176	549.27	73.86	−9.409	298.89	6.180	−1.453
3	−1184.15	72.77	20.53	−637.61	6.028	3.173	548.71	73.71	−9.392	298.73	6.173	−1.451
4	−1182.56	72.57	20.49	−637.14	6.019	3.169	547.96	73.50	−9.369	298.51	6.164	−1.449
5	−1180.57	72.32	20.42	−636.57	6.008	3.164	547.03	73.25	−9.341	298.23	6.152	−1.447
10	−1164.90	70.37	19.94	−631.99	5.920	3.124	539.67	71.24	−9.119	296.03	6.060	−1.428
15	−1140.31	67.32	19.19	−624.69	5.781	3.060	528.14	68.12	−8.771	292.53	5.914	−1.399
20	−1107.95	63.36	18.20	−614.86	5.595	2.975	512.97	64.07	−8.314	287.82	5.718	−1.358
25	−1069.25	59.67	17.02	−602.74	5.367	2.869	494.85	59.29	−7.766	282.02	5.480	−1.309
α [b]	−1186.55	73.07	20.61	−638.30	6.041	3.179	549.84	74.02	−9.426	299.06	6.187	−1.454
$\beta_1 / 10^{-3}$ [b]	200.108	−25.40	−6.134	57.946	−1.117	−0.5063	−94.032	−25.80	2.836	−27.851	−1.167	0.2414
$\beta_2 / 10^{-7}$ [b]	−302.345	73.55	9.558	−49.731	1.225	0.448	145.16	48.31	−4.346	25.090	1.225	−0.276

[a] See footnote a of Table 2.
[b] Coefficients in the fit function for the J dependence: $\alpha + \beta_1 J(J+1) + \beta_2 J^2(J+1)^2$.

Table 12
Calculated magnetic hyperfine constants (M^C, $M^{H/D}$ and D) and the nuclear quadrupole coupling constants $eQ_{C/D}q^{C/D}_{v,J}/h$ for CH^+ in the $v = 0$, $J = 1$ rovibrational state (all values in kHz) [a]

	$eQ_C q^C_{v,J}/h$ [b]	$eQ_D q^D_{v,J}/h$ [b]	M^C	$M^{H/D}$	D
$^{13}CH^+$			−1186.15	73.02	20.59
$^{13}CD^+$		121.84	−638.18	6.039	3.178
$^{11}CH^+$	−15390		549.65	73.97	−9.420
$^{11}CD^+$	−15412	121.84	299.00	6.185	−1.454

[a] See footnote a of Table 2.
[b] The quadrupole coupling constants are taken from Ref. [13].

The comparison of the magnetic hyperfine structure parameters with their quadrupole coupling counterparts [13], in Table 12, shows that one will obtain for each isotopic variant of CH^+ quite a different coupling scheme. Due to the absence of a quadrupole nucleus in $^{13}CH^+$, its hyperfine structure will be governed by the spin–rotation interaction of the C nucleus, perturbed by the much smaller interaction of the H magnetic dipole and the direct nuclear spin–spin interaction. As shown in Tables 10 and 11, the $M^C_{v,J}$ constants are sufficiently large so that the corresponding splitting ($M^C I_C J \approx 2000\ J$ kHz) will be observable at all vibrational lines considered here. In the $^{13}CD^+$ isotopic species the deuterium quadrupole coupling and magnetic interactions are of comparable strength and a perturbation treatment is no longer possible. The level splitting can only be obtained from a solution of the secular equations corresponding to the hyperfine structure Hamiltonian [15]. Finally, it can be seen from Table 12, that the ^{11}C quadruple coupling dominates the hyperfine structure splitting in both radioactive isotopic species.

For the BH molecule, the hyperfine structure parameters of the four isotopomers ^{10}BH, ^{11}BH, ^{10}BD, and ^{11}BD were investigated. The rovibrational dependence of the magnetic coupling constants corresponding to the recently published nuclear quadrupole coupling constants [11] are presented in Tables 13 and 14. It can be seen, that similar to the CH^+ molecule the absolute values of the spin–rotation constants and of the direct nuclear spin–spin interaction constants in all isotopomers of BH decrease with increasing vibrational quantum number v. The vibrational dependence of M^B is much more pronounced than that of $M^{H/D}$. The dependence on the rotational quantum number J is again very weak.

Table 15 collects both the magnetic and the electric constants of the BH isotopomers for the ground vibrational and first rotational state. Despite the very large values of the boron spin–rotation constants, the hyperfine structure of the rovibrational levels of all BH isotopomers will be dominated by the boron quadrupole coupling. This corresponds to the most typical case occurring in the hyperfine spectra of $^1\Sigma$ molecules, i.e. to the coupling of one strong quadrupole interaction with weak magnetic interactions. It results in an additional split of the hyperfine structure levels arising from the quadrupole coupling [15]. Therefore, it is often the case that the relatively small magnetic interactions are observable [60].

Generally, it can be said that although the electronic correlation and rovibrational corrections re-

Table 13
Vibrational dependence of the spin–rotation constants, M^B and $M^{H/D}$, and the direct spin–spin coupling constants, D, for some BH isotopomers in the $J = 1$ rotational state calculated at the CCSDPPA level (all values in kHz) [a]

v	^{11}BH			^{11}BD			^{10}BH			^{10}BD		
	M^B	M^H	D	M^B	M^D	D	M^B	M^H	D	M^B	M^D	D
0	−453.74	13.70	20.20	−247.40	1.138	3.117	−153.19	13.82	6.765	−84.11	1.157	1.044
1	−436.27	13.66	19.45	−240.41	1.138	3.033	−147.27	13.78	6.512	−81.72	1.115	1.015
2	−419.16	13.55	18.65	−233.48	1.136	2.947	−141.49	13.65	6.265	−79.34	1.153	0.986
3	−402.71	13.37	17.93	−226.69	1.131	2.861	−135.93	13.46	6.001	−77.02	1.148	0.957
4	−387.37	13.16	17.20	−220.07	1.122	2.775	−130.75	13.24	5.755	−74.76	1.140	0.928
5	−373.01	12.92	16.49	−213.80	1.113	2.691	−125.90	12.98	5.518	−72.61	1.130	0.900

[a] see footnote a of Table 2.

Table 14
Rotational dependence of M^B, $M^{H/D}$ and D of some BH isotopomers in the ground vibrational state calculated at the CCSDPPA level (all values in kHz) [a]

J	^{11}BH			^{11}BD			^{10}BH			^{10}BD		
	M^B	M^H	D	M^B	M^D	D	M^B	M^H	D	M^B	M^D	D
1	−453.74	13.70	20.20	−247.40	1.138	3.117	−153.19	13.82	6.765	−84.11	1.157	1.044
2	−453.21	13.67	20.18	−247.25	1.136	3.115	−153.01	13.79	6.757	−84.06	1.156	1.043
3	−452.42	13.63	20.14	−247.01	1.134	3.112	−152.74	13.75	6.744	−83.98	1.154	1.042
4	−451.37	13.57	20.09	−246.70	1.132	3.108	−152.39	13.69	6.727	−83.87	1.151	1.041
5	−450.06	13.50	20.03	−246.31	1.129	3.103	−151.94	13.62	6.706	−83.74	1.148	1.039
10	−439.74	12.96	19.54	−243.23	1.104	3.061	−148.43	13.06	6.540	−82.67	1.122	1.025
15	−423.54	12.13	18.76	−238.32	1.064	2.995	−142.92	12.22	6.280	−80.98	1.082	1.002
20	−402.23	11.09	17.76	−231.71	1.012	2.906	−135.67	11.16	5.939	−78.70	1.029	0.972
25	−376.74	9.91	16.55	−223.56	0.951	2.797	−127.00	9.97	5.534	−75.89	0.965	0.935
α [b]	−454.00	13.71	20.21	−247.48	1.138	3.118	153.28	13.83	6.769	−84.14	1.157	1.044
$\beta_1 / 10^{-3}$ [b]	131.78	−6.999	−6.258	38.968	−0.3190	−0.5332	44.821	−7.289	−2.123	13.56	−0.4901	−0.1769
$\beta_2 / 10^{-7}$ [b]	−198.93	17.77	9.483	−33.453	0.470	0.582	−67.578	20.61	3.420	−13.22	2.867	0.134

[a] See footnote a of Table 2.
[b] Coefficients in the fit function for the J dependence: $\alpha + \beta_1 J(J+1) + \beta_2 J^2(J+1)^2$.

Table 15
Calculated magnetic hyperfine constants (M^B,$M^{H/D}$ and D) and the nuclear quadrupole coupling constants $eQ_{B/D}q^{B/D}_{v,J}/h$ for BH in the $v=0$, $J=1$ rovibrational state (all values in kHz) [a]

	$eQ_B q^B_{v,J}/h$ [b]	$eQ_D q^D_{v,J}/h$ [b]	M^B	$M^{H/D}$	D
[11]BH	−6595.1		−453.74	13.70	20.20
[11]BD	−6618.9	111.41	−247.40	1.138	3.117
[11]BH	−13744.0		−153.19	13.80	6.765
[10]BD	−13794.0	111.40	−84.11	1.157	1.044

[a] See footnote a of Table 2.
[b] The quadrupole coupling constants are taken from Ref. [11].

duce the very large values of the spin–rotation constants M^C and M^B in CH^+ and in BH, as predicted by earlier CHF calculations [33], they leave them still large enough to produce a significant effect on the hyperfine structure patterns of the rotational spectra for all the isotopic species studied here.

4. Conclusions

Spin–rotation constants were calculated for the six lowest vibrational states of some isotopomers of LiH, HF, CH^+ and BH. It was found, that, within the Born–Oppenheimer separation of electronic and nuclear motion, reliable data can be obtained if the electronic calculations are based on correlated polarization propagator approximations using extended basis sets of atomic orbitals adapted for magnetic properties. The rovibrational dependence of the spin–rotation constants was found to be less sensitive to the level of correlation than their absolute values. Good agreement with experimental data for the isotopomers of LiH and HF encouraged us to predict values for BH and CH^+. An improved semi-experimental value for the nuclear magnetic shielding constants of HF was derived from recently measured spin–rotation constants and our calculated diamagnetic contribution. The hyperfine splitting patterns for the various isotopomers were discussed.

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